

WORKSHOP: EMPLOYEE MONITORING AND RECRUITMENT

(1)

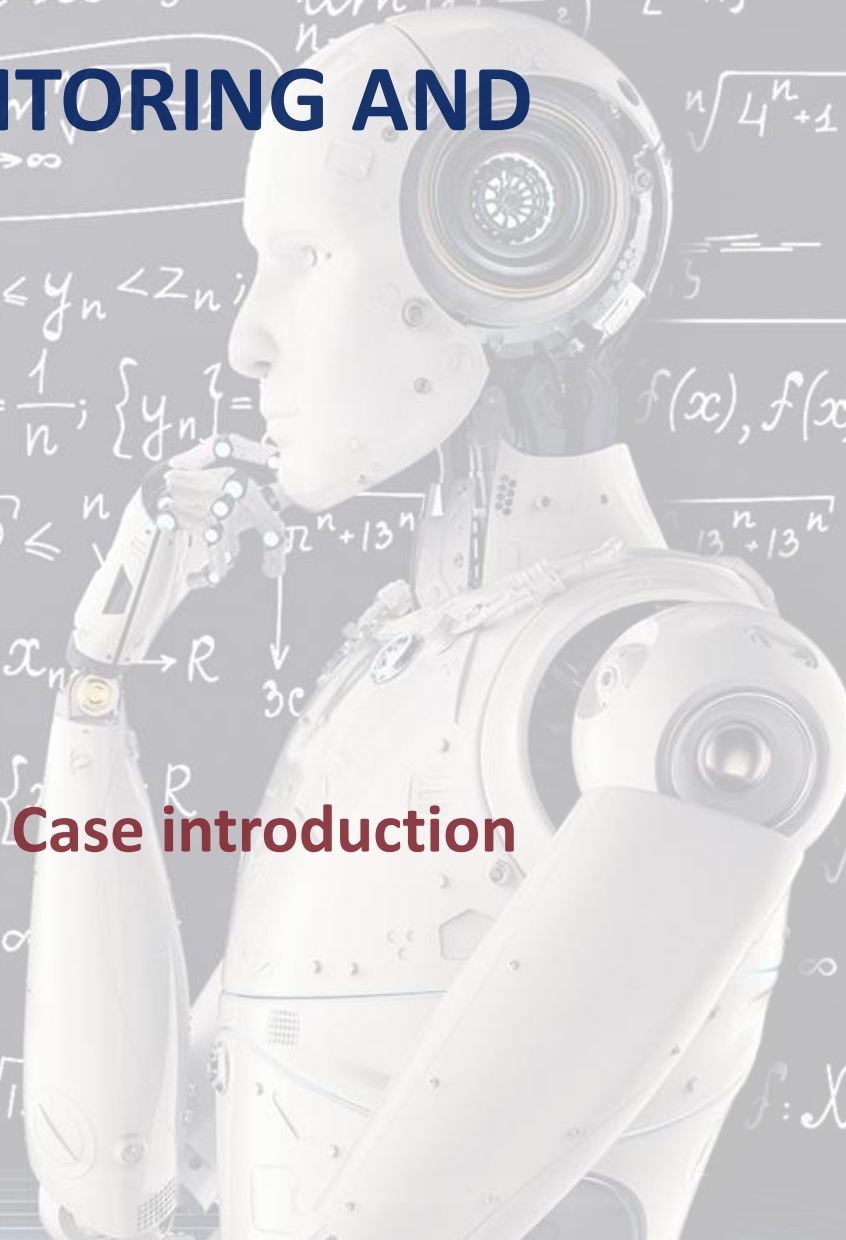
Module 3

Candidate Matching

Lesson 1

Search&Match Theory and Matching Use Case introduction

Milan, 18th March 2026



A recap on recruitment process

the recruitment process is a multi-step process. To summarize it we can think about two different scenarios, which usually happen simultaneously: a candidate is looking for a job and a company needs a resource to strengthen its operations.

The recruiter's job is to be the intermediary in this negotiation, selecting the most suitable candidate for a job position or, conversely, finding the right job offer for a candidate.

Although there can be direct contacts between recruiters and candidates (for example Adecco local branches), there are also job posts that recruiters can write to present a job offer. Needless to say, job posts need to be well written and attractive to be considered by a larger pool of candidates



GPT Job Posting and GPT Matching – How GenAI can support the recruitment process

Generative AI can be useful to support the recruiter in his process to select the right candidate or to offer the most suitable job. In Adecco two examples of how such technology is used in this context are GPT Job Posting and GPT Matching.

As the software's names imply, both leverage a GPT model to perform their tasks;

- GPT Job Posting: uses Azure OpenAI GPT to generate job posts based on recruiters' inputs. Specific input parameters are provided instead of an open text solution to guide the user in the creation process and make it easier to use. There are also other tasks that it can perform, such as translating or creating interviews questions - [Demo](#)
- GPT Matching: matching engine that provides correspondences between job offers and candidates (in both directions). It also gives explanations on why such matches are found - [Demo](#)

Matching engine - Problem statement

The objective of this exercise is to develop an algorithm that matches candidates with job positions based on the analysis of textual data. Candidates will have access to anonymized resumes and job descriptions.

Key Requirements

- **Text Analysis:** Implement a method to process and extract relevant information from both resumes and job postings.
- **Recommendation Engine:** Design a matching system that identifies the most suitable job positions for each candidate.
- **Affinity Scoring:** Assign a similarity score to each candidate-job pair, ranking matches based on relevance.

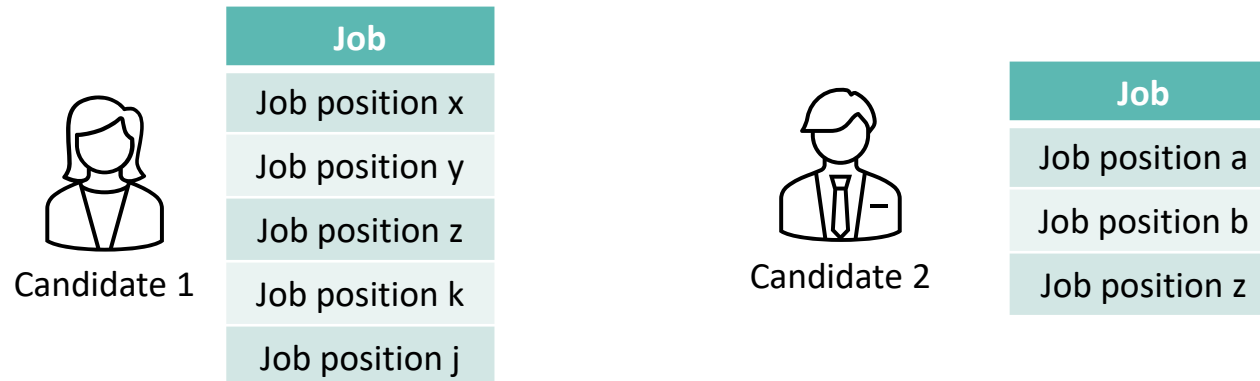
There are no constraints on the design of the solution; however, it should include a structured workflow for text analysis and recommendation generation. The implementation will be developed in **Python**, leveraging **an AI-based approach** of your choice to analyze and compare textual data effectively.

Dataset description – Candidates and Jobs

The data provided to carry out the exercise is composed by anonymized candidates CVs and job posts. Each CV and job post are identified using, respectively, «Candidate_ID» and «Offer_ID».

Each candidate is linked (i.e., it fits) to at least one job position and up to a maximum of five.

Thus, each candidate is associated with several job positions but also one job position can be associated with several candidates.



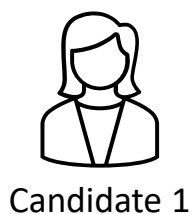
Dataset description – Ranking

Each candidate-job association is assigned a score that represents the degree of affinity between the candidate and the specific role.

The ranking system relies on these weights, allowing distinctions between strong and latent matches. A high score indicates a strong affinity between the candidate and the position, suggesting a higher suitability for that role.

Conversely, a lower weight signals a weaker but still relevant association.

This approach models real-world scenarios where candidates may be more or less suited for a position.



Job	Rank	Score
Job position x	1	14
Job position y	2	12
Job position z	3	7
Job position k	4	6
Job position j	5	5



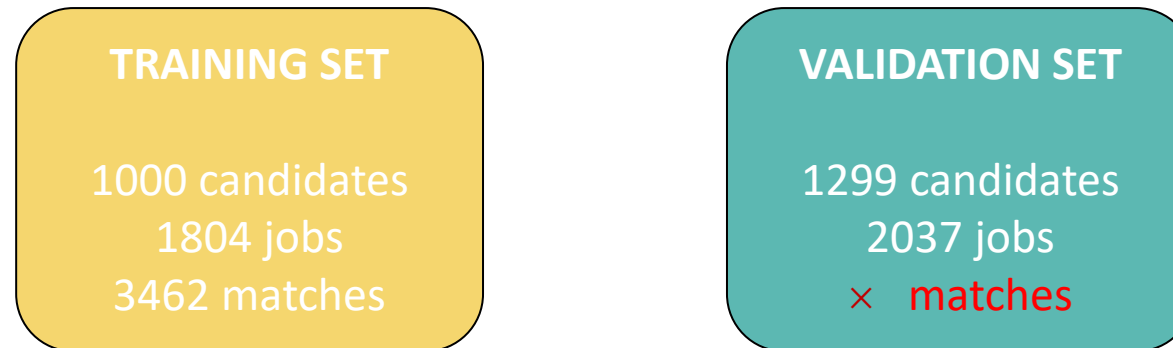
Job	Rank	Score
Job position a	1	14
Job position b	2	12
Job position z	3	7

Dataset description – Training and Validation

The initial dataset has been divided into a **training set** and a **validation set**, each serving a distinct purpose in the evaluation process.

The **training set** includes a selection of candidates with their CVs, along with a subset of job positions that contain job titles and full job descriptions. It also provides explicit candidate-job associations, complete with rank and score values. In contrast, the **validation set** contains a completely separate group of candidates and a list of job positions, but no predefined associations between them

This division ensures that the algorithm learns from known candidate-job relationships during training and is then tested on unseen data, assessing its ability to generalize and predict suitable job matches.



Matching algorithm – Expected outcome

The expected result includes an additional sheet in the file containing the validation set. This sheet should list all possible candidate-job position combinations, assigning an affinity score to each pair.

The primary focus is not on the exact numerical value of the score but on the **ranking** it establishes. The goal is to correctly order the job positions for each candidate based on their predicted suitability, ensuring that the most relevant matches appear at the top of the list.

The submission must include the code for:

CV text analyzer

Job post analyzer

**Recommendation
engine**